

Task 3 Report: Fairness, Bias, and Explainability Analysis

Objective

The objective of this task was to analyze a machine learning model from the perspective of fairness, bias, and explainability. The project focused on understanding how the model makes decisions, whether its behavior differs across sensitive groups, and what mitigation steps can be proposed to reduce bias.scribd+1

Dataset

A customer churn dataset was used for this analysis. The dataset contained customer-related features such as Gender, Age, Salary, ContractType, MonthlyCharges, and TotalCharges. The **Gender** attribute was treated as the sensitive group for fairness analysis.[github](#)

Model Used

A **Random Forest Classifier** was trained on the churn dataset. This model was chosen because it performs well on classification tasks and provides global feature importance values, which are useful for interpretation.[github](#)

Explainability Method

Explainability was performed using **feature importance** and **SHAP**. Feature importance was used to understand which variables had the strongest influence on the churn prediction, while SHAP was used for both global and local explanation to show how each feature contributed to the model output. SHAP is specifically designed to explain model outputs using Shapley-value-based local explanations.[scribd](#)

Bias Check Results

The model's predictions were compared across the Gender groups. The analysis included total samples per group, actual positive churn rate per group, predicted positive churn rate per group, and group-wise accuracy. This comparison helped identify whether the model showed different decision patterns for different gender groups. If the predicted positive rate differs substantially across groups, that may indicate a fairness concern that should be investigated further.[github](#)

Mitigation Recommendations

Based on the fairness analysis, the following mitigation actions are recommended:

1. Monitor model predictions across sensitive groups on a regular basis.
2. Review whether the sensitive attribute should be retained, transformed, or removed.
3. Tune decision thresholds if one group is affected more strongly than others.
4. Collect more balanced and representative data.
5. Use fairness checks and SHAP explanations together before model deployment.scribd+1

Conclusion

This task demonstrated that model evaluation should go beyond accuracy. By combining feature importance, SHAP explanations, and group-wise fairness checks, it becomes easier to understand model behavior and propose practical mitigation strategies.



